System Analysis Work Sample

Project: Government Counter Appointment Booking System

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Disclaimer: Prepared as an illustrative case study for demonstration purposes. This is a hypothetical project and is not based on an actual SDOC assignment or a completed government implementation. No stakeholder interviews, field research, implementation or user testing have been carried out.

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1. Executive Summary

Many government offices serve citizens at counters on a first-come, first-served basis. For this case study, it is assumed that citizens may have difficulty predicting waiting times and that staff have limited visibility into expected demand. These are problem assumptions for the exercise, not findings from measured data or field research.

The proposed solution is a web-based appointment booking system. Citizens can reserve a time slot for a counter service, receive a reference number and SMS confirmation, and attend at the scheduled time. Citizens without internet access can receive assistance from staff to make a booking. Staff can track each appointment through five statuses: Confirmed, Checked-in, Completed, Cancelled, and No-show. An administrator can manage slot capacity and office closures. The intended users are citizens, counter staff, the office head, and system administrators.

The system is designed to support more predictable visits, reduce unmanaged queuing, prevent bookings from exceeding configured slot capacity, and provide staff with a daily schedule and appointment-status records. These are intended outcomes, not proven results. Their actual impact would need to be evaluated using operational data after implementation.

2. Problem Statement and Scope

2.1 Problem Statement

Citizens may attend a counter without a reserved time and wait in a first-come, first-served queue. In this case study, it is assumed that staff have limited advance visibility into expected demand and that attendance records may depend on the office's existing process. These assumptions have not been verified through field research.

A booking system could help organize appointments, but it must also accommodate citizens who lack internet access and prevent bookings from exceeding the available capacity of a time slot.

2.2 Objectives

- Let citizens reserve a slot for a specific service in advance and quote a reference number at the counter.
- Prevent bookings from exceeding configured slot capacity, including when simultaneous booking requests are received.
- Give staff a daily schedule and a clear status for every appointment.
- Handle cancellation, rescheduling and office closures without leaving citizens uninformed.
- Provide a staff-assisted booking route for citizens without internet access and define how walk-ins will be handled.

2.3 Scope

Table 2.3.1 Scope of the proposed system

Included	Excluded
• Online booking, cancellation and rescheduling by citizens • SMS confirmation and cancellation notices • Staff-assisted booking • Appointment Check-in and status updates by counter staff • Administrator maintenance of slots, capacity and closures • Daily schedule and audit log	• Payment of service fees • Document upload or verification • Integration with other government systems • Live queue display or waiting-time estimates • A dedicated mobile application • Analytics dashboards and multi-office roll-out

2.4 Constraints

- No budget, hosting environment or technology stack has been defined; therefore, this document does not prescribe a specific technology.
- The proposed system must fit the existing counter workflow and staff capabilities, with a documented manual fallback for service interruptions.
- Applicable government IT security, privacy and data-protection requirements have not yet been identified. The relevant requirements must be confirmed with the responsible authority before implementation (Q-08).

2.5 Assumptions

Eight working assumptions (A-01 to A-08) underpin this analysis. All are unconfirmed; Table 5 in Section 7.1 identifies the proposed validator for each assumption.

3. Stakeholders and Proposed Workflow

3.1 Stakeholders

Table 3.1.1 Stakeholders and their needs

Stakeholder	Role	Key needs
Citizen	Requests the counter service; books online or at the counter	Avoid unnecessary trips and uncertain waiting times; receive clear confirmation; cancel or reschedule easily; use Nepali-language pages and messages; access staff-assisted booking without internet access.
Counter staff	Serve citizens, check them in and update status	Accurate daily schedule; quick lookup by reference or mobile number; clear status; a practical way to handle walk-ins and late arrivals.
Office head (service owner)	Decides service policy and capacity	Control over slots, closures, walk-in and cancellation rules; assurance that the system is fair to all citizens.
System administrator	Responsible for system operation, access management and technical support, the responsible organization and role allocation remain to be confirmed.	Secure access; backups; visibility of failed notifications and the audit log; a manageable support workload.
SMS provider (external)	Delivers notifications	Not a user; a dependency whose availability and cost affect FR-06 and FR-14.

3.2 Proposed Workflow

Figure 1 illustrates the proposed workflow from booking request to final appointment outcome. The system revalidates slot availability at confirmation time (FR-03), because another request may reserve the remaining capacity after the citizen initially views the slot.

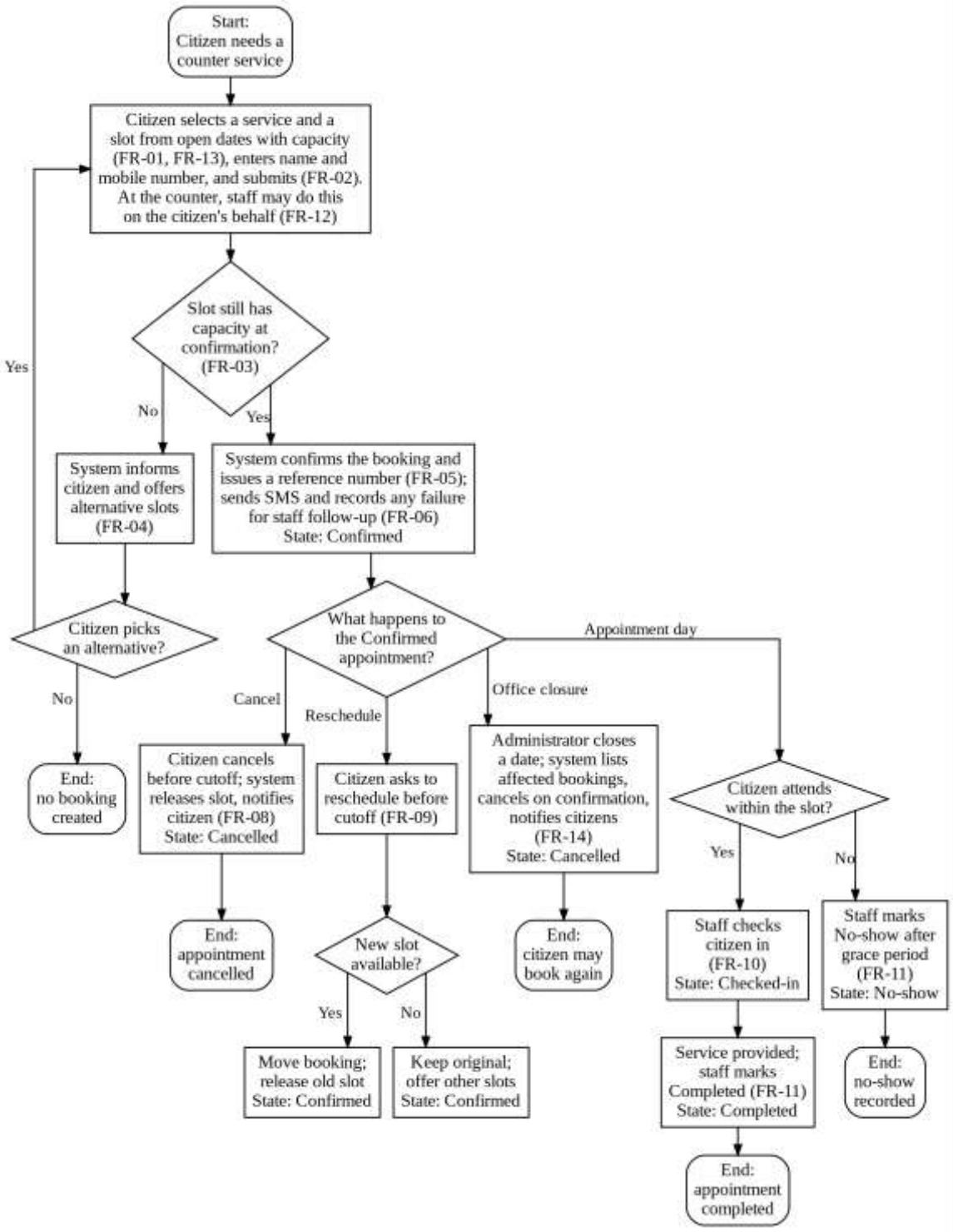


Figure 3.2.1 Proposed Appointment Booking Workflow

Four alternative paths need particular attention:

- Unavailable slot: No booking is created. The citizen is offered alternative slots and may choose one or stop (FR-04).

- Rescheduling: Treated as a change to the existing appointment. The system validates and reserves the new slot while releasing the original slot as one atomic operation. If the new slot is unavailable, the original appointment remains unchanged (FR-09). The appointment status remains Confirmed, and the audit log records the original and new slots.
- Office closure: The system identifies appointments affected by a closure. The proposed policy is to cancel those appointments with the reason “office closure” and notify affected citizens so they can make a new booking (FR-14). The cancellation and notification policy requires validation by the service owner.
- Walk-ins and no internet access: Staff can create appointments for citizens who cannot book online, subject to the same slot-capacity rules (FR-12). Whether citizens without appointments may also be served as walk-ins, and whether capacity should be reserved for them, remain open policy decisions (Q-02).

3.3 Appointment Status Model

Figure 2 illustrates the permitted appointment-status transitions. Completed, Cancelled and No-show are terminal states under the proposed model. The system rejects transitions that are not explicitly permitted (FR-11). The handling of a citizen who checks in but leaves before receiving the service is unresolved because the current five-status model does not represent this outcome (Q-05).

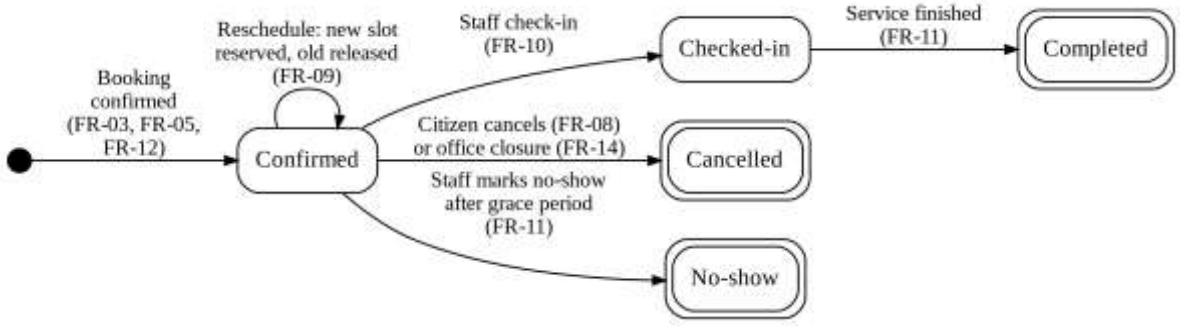


Figure 3.3.1 Appointment State Diagram

4. Functional Requirements

Each requirement describes one testable behaviour. Values in square brackets are configurable parameters that depend on the decisions in Section 7.3. The acceptance criteria in Section 6 use the same IDs.

Table 4.1 Functional requirements

ID	Requirement
FR-01	The system must display, for a selected service, only future dates and time slots that fall on open days and have remaining capacity.
FR-02	The system shall require a service, a slot, the citizen's full name and a mobile number in a valid format before a booking request can be submitted.
FR-03	The system shall confirm a booking only if the selected slot still has remaining capacity at the moment of confirmation, and shall ensure that simultaneous requests can never produce more confirmed bookings than the slot's capacity.
FR-04	If the selected slot has no remaining capacity at confirmation, the system shall create no booking, inform the citizen, and offer the nearest available alternative slots for the same service.
FR-05	The system shall assign each confirmed booking a unique reference number, set its status to Confirmed, and show the service, date, time, office and reference number to the citizen.
FR-06	The system shall send an SMS confirmation to the citizen's mobile number. If sending fails, the booking shall remain Confirmed, and the failure shall be recorded and listed for staff follow-up.
FR-07	The system shall prevent a citizen, identified by registered mobile number, from holding more than [maximum] Confirmed appointment(s) for the same service within [defined period or scope].
FR-08	The system shall allow a citizen who provides the appointment reference number and completes the approved verification process to cancel a Confirmed appointment until [cutoff period] before its scheduled start time. On successful cancellation, the system shall release the slot capacity and attempt to notify the citizen.
FR-09	Under the same conditions as FR-08, the system shall allow a citizen to reschedule a Confirmed appointment to another available slot of the same service. The new slot shall be reserved and the original slot released as a single operation; if the new slot is unavailable, the original appointment shall remain unchanged.
FR-10	The system shall allow counter staff to find an appointment by reference number or mobile number and record check-in, changing its status from Confirmed to Checked-in.
FR-11	The system shall allow counter staff to change an appointment status from Checked-in to Completed, and from Confirmed to No-show only after the scheduled start time plus [grace period] has passed. The system shall reject all other status transitions not permitted by Figure 2.
FR-12	The system shall allow authorized counter staff to create an appointment on behalf of a citizen, subject to FR-03. Whether unbooked walk-ins may be accepted, and under what capacity rules, shall be governed by the configured [walk-in policy].
FR-13	The system shall allow an authorised administrator to define open days, closure dates, slot times and slot capacity for each service, and shall reject a capacity lower than the number of Confirmed appointments already in the slot.
FR-14	When an administrator closes a date or period that contains Confirmed appointments, the system shall list them and, on the administrator's confirmation, set them to Cancelled with the reason "office closure", release their capacity, and attempt to notify the affected citizens, listing any failed notifications for staff follow-up.

ID	Requirement
FR-15	The system shall provide counter staff with a daily schedule showing, for each slot, every appointment with its reference number, service and current status.
FR-16	The system shall record every booking, cancellation, rescheduling and status change with the date, time, acting user (or “citizen”), and relevant previous and new status. For rescheduling, the log shall also record the original appointment slot and the new appointment slot. Staff users shall not be permitted to edit or delete audit-log records.

5. Non-Functional Requirements

Every value below is a proposed target for discussion. None has been confirmed with a stakeholder, and none is presented as a legal or standards requirement.

Table 5.1 Proposed non-functional requirements

ID	Area	Proposed requirement	Validate with
NFR-01	Performance	For 95% of requests under a proposed peak load of [100] simultaneous users, the slot list shall display within 3 seconds and booking confirmation shall complete within 5 seconds. These targets shall be validated through performance testing under an agreed test environment and workload.	Office head (expected demand)
NFR-02	Availability	Booking and staff functions should be available during defined office hours, with a proposed target of 99% monthly availability, excluding agreed announced maintenance. During an outage, staff shall follow the documented manual-register fallback procedure. The availability measurement method and outage-handling procedure remain to be agreed.	Office head, SDOC
NFR-03	Security	Permissions shall restrict each staff role to its authorized functions, and the applicable government security requirements shall be identified and validated before implementation.	SDOC
NFR-04	Privacy	The system shall collect only the personal data necessary to operate the booking process. The authority responsible shall define and approve data-access rules and retention periods before implementation.	SDOC, records authority
NFR-05	Accessibility and usability	Citizen-facing pages shall support commonly used low-cost smartphone browsers, use plain language, and aim to complete the booking process in no more than [5] user steps. A staff-assisted booking route shall be available for citizens without internet access (FR-12). The applicable accessibility criteria shall be confirmed with the responsible authority.	Office head, SDOC
NFR-06	Language	Pages and SMS text shall be available in Nepali and English, and names entered in Nepali script shall be accepted. The date calendar(s) displayed are to be confirmed.	Office head, SDOC
NFR-07	Recovery	Booking data shall be backed up regularly and restorable. Recovery point and recovery time targets of [to be set] shall be agreed, and a restore shall be tested before go-live.	SDOC
NFR-08	Operational support	A named support contact shall exist. Administrators shall be able to view failed notifications and the audit log. A short user guide and staff training shall be provided before go-live.	SDOC, office head

6. Acceptance Criteria and Traceability

Figure 3 links each functional requirement to a measurable acceptance criterion, the figure in which its behaviour appears, and a proposed priority (M = Must, S = Should). Priorities are proposals to be confirmed by the office head. Criteria for the non-functional requirements would be written once their targets are confirmed. The concurrency scenario is covered under FR-03.

Table 6.1 Requirements Traceability Table

ID	Acceptance criterion	Shown in	Priority
FR-01	Given a service with three open future slots, one full slot and one closed date, the system displays only the three open slots with remaining capacity. The full slot and closed date are not displayed.	Fig 1	M
FR-02	A request with a missing name, a missing mobile number or a number failing the agreed format check is refused with a message naming the field; no booking is created. A valid request proceeds to confirmation.	Fig 1	M
FR-03	Concurrency scenario: a slot has exactly one place left and two citizens submit confirmation within the same second (automated test, at least 100 runs). In every run exactly one booking is Confirmed, the other is refused, and the slot's confirmed count never exceeds its capacity.	Fig 1, 2	M
FR-04	In the scenario above, the refused citizen sees a "slot no longer available" message and up to [3] nearest alternative slots (or a message that none exist). No booking record exists for the refused request.	Fig 1	M
FR-05	Each confirmed booking has status Confirmed and a reference number unique across at least 1,000 test bookings. The confirmation shows service, date, time, office and reference number.	Fig 1, 2	M
FR-06	Following booking confirmation, the system attempts to send an SMS containing the reference number, date and time. When the SMS service is unavailable during testing, the booking remains Confirmed, the failure is recorded, and it appears on the staff follow-up list.	Fig 1	S
FR-07	With the limit at [1], a second booking for the same service and mobile number is refused while the first is Confirmed, and the message shows the existing date. It is accepted once the first is no longer Confirmed.	—	S
FR-08	Cancelling before the cutoff sets Cancelled, the slot reappears in FR-01 results immediately, and the system attempts to send an SMS. A wrong mobile number, or a request after the cutoff, is refused and the status is unchanged.	Fig 1, 2	M
FR-09	Rescheduling to an available slot retains the appointment reference number and Confirmed status, releases capacity in the original slot and reserves capacity in the new slot. If the new slot becomes unavailable before confirmation, the request is refused and the original appointment remains unchanged. The audit log records the original slot, new slot, timestamp and acting user. Verify that capacity is not exceeded in either slot.	Fig 1, 2	S
FR-10	Searching by reference or mobile number returns the matching Confirmed appointment; check-in sets Checked-in and records the time. Check-in on a Cancelled, No-show or Completed appointment is refused.	Fig 1, 2	M

ID	Acceptance criterion	Shown in	Priority
FR-11	Completed can be set only from Checked-in, and No-show only from Confirmed after start time plus grace period. Any transition not in Figure 2 (for example Completed to Cancelled) is refused and the status is unchanged.	Fig 1, 2	M
FR-12	Staff can create a booking for a citizen at the counter; a booking into a full slot is refused. The appointment appears in the same schedule and follows the same statuses. Walk-in acceptance follows the decision on Q-02.	Fig 1, 2	M
FR-13	After a date is marked closed, no slots for it are offered under FR-01. Reducing a slot's capacity below its number of Confirmed appointments is refused.	Fig 1	M
FR-14	Closing a date with five Confirmed test appointments lists all five. After the administrator confirms, all five become Cancelled with reason "office closure," capacity is released, and the system attempts to notify all five affected citizens. Any failed SMS appears on the staff follow-up list.	Fig 1, 2	M
FR-15	The schedule for a chosen date lists every appointment by slot with reference number, service and status. A status change by one staff user is visible to another after a page refresh.	—	M
FR-16	After a test sequence of booking, rescheduling, cancellation, check-in and completion, the log holds one entry per action with timestamp, actor, and previous and new status. Staff cannot edit or delete entries.	—	M

7. Risks, Assumptions, and Open Questions

7.1 Assumptions

The assumptions, risks, and open questions below are provisional and have not been validated through stakeholder interviews or field research. They are included to identify decisions and dependencies that would need to be addressed before implementation.

Table 7.1.1 Assumptions (all unconfirmed)

ID	Assumption	Validate with
A-01	Each service is offered at one office location with a fixed number of service positions per time slot (the slot capacity).	Office head, counter staff
A-02	Each service has a fixed appointment duration that the office can state.	Counter staff (sample of transactions)
A-03	Citizens are identified at booking by full name and mobile number; identity documents are checked at the counter, not online.	Office head
A-04	SMS can be sent to citizens through a provider approved for government use; email is not required.	SDOC, procurement
A-05	Cancellation and rescheduling are allowed until a cutoff time before the appointment.	Office head
A-06	It is provisionally assumed that walk-ins may be accepted when capacity is available. Whether capacity should be reserved specifically for walk-ins remains an open policy decision for the office head.	Office head
A-07	Working days, hours and holidays are known in advance and entered by an administrator.	Office head, administrator
A-08	No automatic penalty is applied for a No-show.	Office head

7.2 Risks

Table 7.2.1 Risks and proposed responses

ID	Risk	Proposed response
R-01	Slot durations or capacity are set wrongly, so queues continue or counters sit idle.	Check real service times with counter staff before fixing slot lengths; keep duration and capacity adjustable (FR-13); review after initial use.
R-02	Citizens without internet access or a mobile phone are disadvantaged.	Keep staff-assisted booking (FR-12), confirm the walk-in policy (Q-02), and provide an alternative process for citizens who cannot access or use a mobile phone.
R-03	SMS is unreliable or unavailable.	Show the confirmation on screen, keep the reference retrievable at the counter, and record failed notification attempts for follow-up (FR-06, FR-14). Confirm the provider (Q-06).
R-04	Citizens book and do not attend, wasting capacity.	Record No-show (FR-11) so the extent can be measured. Decide any policy response only when data exists; no penalty is assumed (A-08).

ID	Risk	Proposed response
R-05	Staff keep using the manual routine, so system records diverge from reality.	Involve counter staff in requirements review, provide training before go-live, and agree on a temporary parallel manual-register period to reduce disruption during transition.
R-06	Names and mobile numbers are held without agreed access and retention rules.	Collect minimum data (NFR-04), apply role-based access (NFR-03), keep the audit log (FR-16), and obtain retention rules (Q-08).
R-07	Stakeholders disagree, for example on walk-ins versus appointments only.	Present options with their consequences and ask the office head to decide (Section 8).

7.3 Open Questions

Table 7.3.1 Decisions needed before implementation

ID	Question	Decision from	Affects
Q-01	Which services and offices are in scope, how many service positions does each slot have, and how long is each service slot?	Office head, counter staff	A-01, A-02, FR-01, FR-03, FR-13
Q-02	Are walk-ins served only when capacity is free, or is part of the capacity reserved for them? How is a citizen who arrives after being marked No-show handled?	Office head	A-06, FR-11, FR-12
Q-03	What are the cancellation and rescheduling cutoff and the limit on active appointments per citizen?	Office head	A-05, FR-07 to FR-09
Q-04	What grace period applies before No-show, and should staff or the system record it?	Office head, counter staff	FR-11
Q-05	Which status applies if a citizen leaves after check-in without being served?	Office head	Figure 2, FR-11
Q-06	Which SMS provider is available, does it support Nepali-script messages, and is email needed?	SDOC, procurement	A-04, FR-06, NFR-06
Q-07	What verification method should be required to prevent unauthorized bookings, cancellations and rescheduling? Should mobile-number verification using a one-time code be implemented, and what alternative should be available to citizens who cannot complete that verification?	SDOC, office head	FR-02, NFR-03
Q-08	Which government security, data-protection and records rules apply, including hosting and retention periods?	SDOC, records authority	NFR-03, NFR-04, NFR-07
Q-09	Which calendar(s) must the interface show, and who maintains the holiday calendar?	Office head, administrator	A-07, FR-13, NFR-06

8. Validation and Next Steps

This is a proposed approach. No stakeholder consultation, field research or testing has been carried out for this case study.

1. Confirm scope and workflow: Walk the office head and counter staff through Section 2 and Figures 1 and 2 and record any corrections.
2. Validate assumptions and open questions: Interview counter staff and the office head and observe a sample of counter transactions to confirm service durations, busy periods and exceptions. Record each decision in a decision log against its A- or Q- reference.
3. Resolve conflicting needs: Where stakeholders disagree, set out the options and their effect on waiting, fairness and staff workload, ask the office head to decide, and record the reason. Escalate to the project sponsor if the disagreement remains.
4. Obtain approval: Prepare a version-controlled requirements baseline with every open item resolved or explicitly deferred. Request sign-off from the office head and technical owner. Manage subsequent changes through change requests that reference requirement IDs and update Section 6 where necessary.
5. Support acceptance testing: Derive test cases from Section 6 and prepare test data covering a full slot, a closed date, and an unavailable SMS service. Coordinate concurrency testing with the technical team, conduct walkthroughs with counter staff, and record defects against requirement IDs until the agreed acceptance criteria are met and sign-off is obtained.